

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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Quarterly Census of Employment and Wages (QCEW).¹⁰ These prior papers use data that lack either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).¹¹ In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this information to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several sample restrictions for our main analysis sample. We include only workers employed by firms that use ADP’s payroll product to maintain worker earnings records. We also exclude employees classified by firms as part-time from the analysis and subset to people between the age of 18 and 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms across our main sample period by keeping only companies that have employee earnings records for each month from January 2021 through July 2025.

In addition, ADP observes job titles for about 70% of workers in its system. We exclude workers who do not have a recorded job title. There are over 7,000 standardized job titles, examples of which include “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of

¹⁰Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

¹¹As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

70% of workers have a college degree according to the 2017 American Community Survey (ACS).²⁵ In Figure A13 we show trends for occupations in which fewer than 30% of workers have a college degree.

Occupations with a high share of college graduates have declining employment overall, with muted differences between more-exposed and less-exposed occupations compared to our main results. In contrast, occupations with a low share of college graduates have rising overall employment, with the least AI-exposed occupations growing and the most exposed occupations declining in employment. Further, for lower college share occupations, the dispersion in employment outcomes is visible in higher age groups as well, with workers up to age 40 showing separation in employment trends by AI exposure. These findings suggest that deteriorating education outcomes cannot fully explain our main results. They also suggest that for non-college workers, experience may serve as less of a buffer to labor market disruption than for college workers.

Other Robustness Checks Figures A14 and A15 show results separately for men and women. The results are similar, suggesting that diverging prospects for men and women are not driving our findings. Figure A16 shows that results are similar when we do not take a balanced sample of firms. Figure A17 shows similar results when including part-time and temporary workers.²⁶

Comparison to CPS Data A useful benchmark for our findings is to compare them to estimates from the monthly Current Population Survey (CPS). The CPS surveys about 60,000 households nationwide each month to collect data on employment and other labor force characteristics. These data are released a few weeks after the reference month, giving close to real-time estimates of employment statistics. A number of prior analyses have used the CPS to assess how AI is impacting entry-level work (Chandar, 2025b; Dominski and Lee, 2025; Lim et al., 2025; Eckhardt and Goldschlag, 2025). We compare some of our main findings in the ADP data to estimates from the CPS.

Figures A18 through A20 show employment changes by age for software developers, customer

²⁵Not a single occupation in the first quintile of GPT-4 β based exposure measure has a college share above 35%, so that quintile is excluded from the results in Figure A12.

²⁶Another possibility is that employment trends are driven by Covid-19 Pandemic-era stimulus checks that distorted labor supply. However, these stimulus payments were conditioned on income requirements, and more AI-exposed occupations on average have higher incomes (Kochhar, 2023), suggesting the observed declines in AI-exposed occupations are unlikely to be driven by this channel.

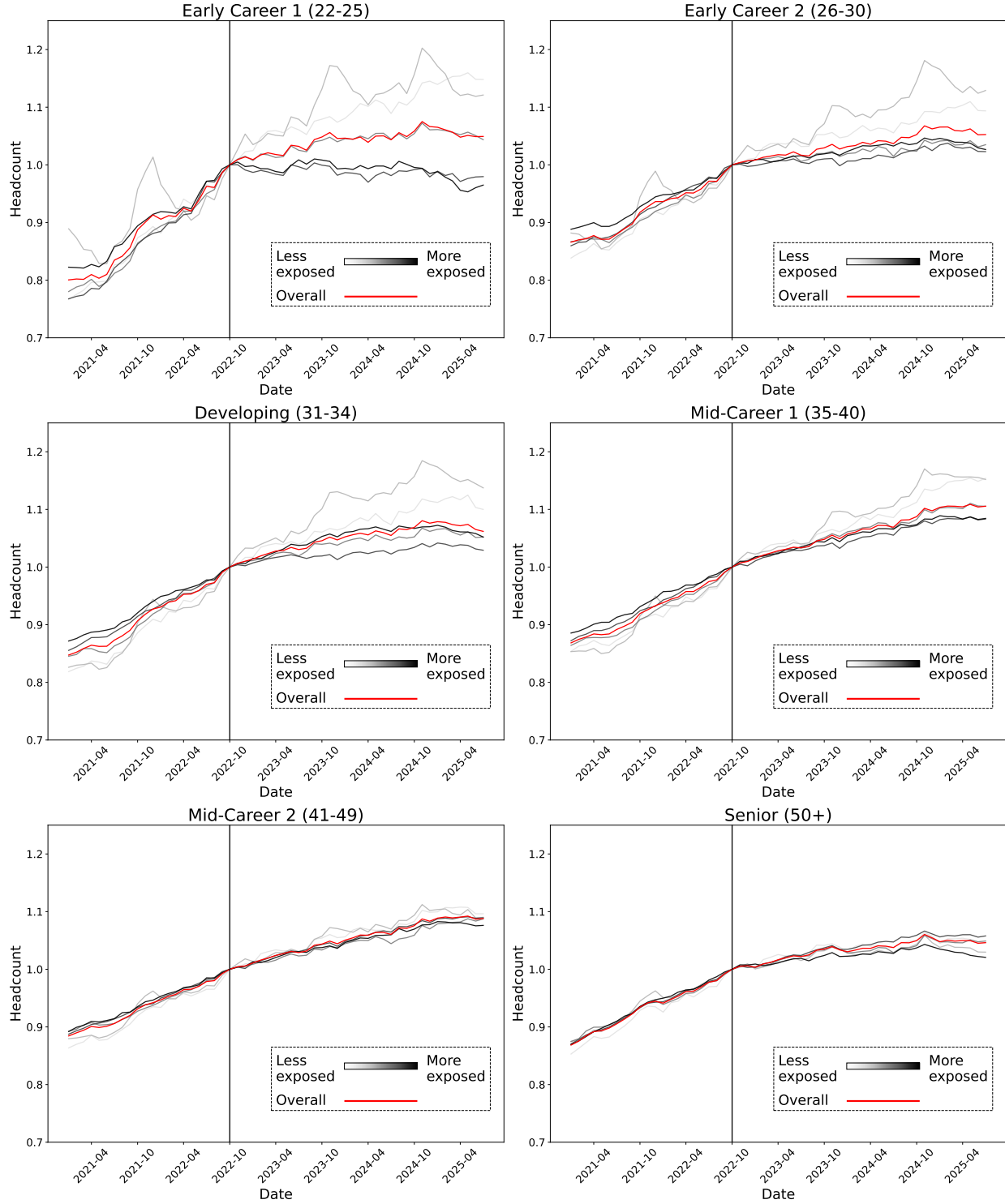


Figure A14: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Considering only men.

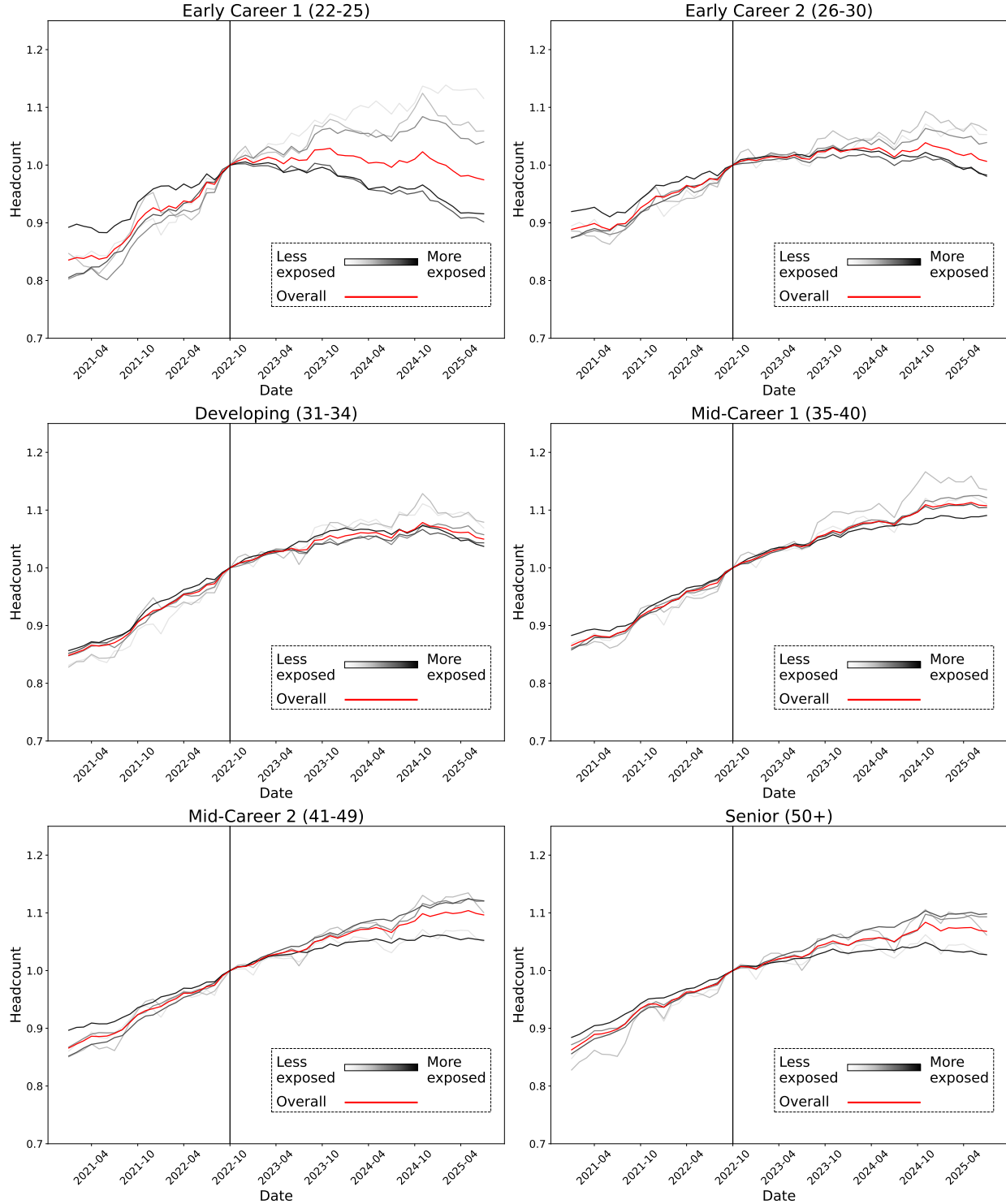


Figure A15: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Considering only women.